Water Quality Report

Annual Drinking Water Quality Summary | Reporting Period: 1 Jan - 31 Dec 2025

Water System	Riverbend Municipal Water Supply (Sample)	**Report Date**	15 January 2026
Source	Surface water (Riverbend Reservoir) + 2 boreholes	**Population Served**	~120,000
Treatment	Coagulation, filtration, chlorination, UV	**Samples Collected**	1,248

1. Executive Summary

Drinking water supplied in 2025 met all applicable health-based standards. Of 1,248 samples analysed, **99.4%** were fully compliant. Seven minor exceedances of the aesthetic turbidity guideline occurred following heavy monsoon rainfall in October; all were corrected within 24 hours and none posed a health risk. No microbiological (E. coli) detections were recorded in treated water leaving the plant or in the distribution network.

Overall Compliance	Microbiological	Chemical	Physical / Aesthetic
99.4%	**100% compliant**	**100% compliant**	**98.6% compliant**

2. Physical and Chemical Parameters

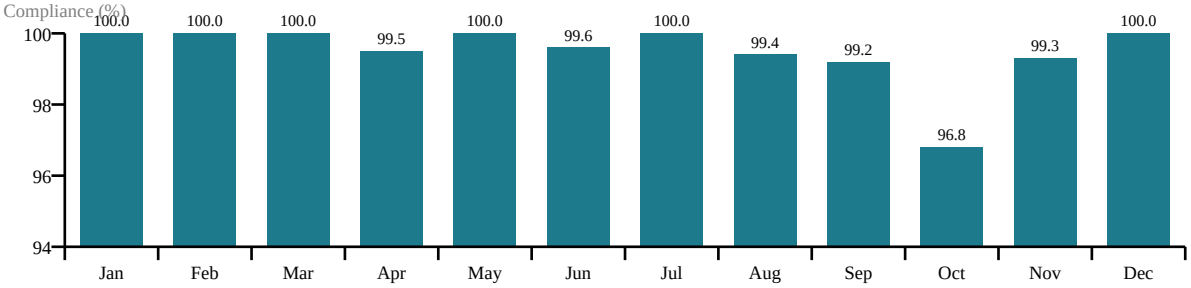
Parameter	Unit	Standard (Limit)	Min	Avg	Max	Status
pH	-	6.5 - 8.5	6.9	7.4	7.9	Compliant
Turbidity	NTU	≤ 1.0 (5.0 max)	0.1	0.4	1.8	Minor exceedance
Total Dissolved Solids	mg/L	500 (2000 max)	210	286	342	Compliant
Total Hardness ($CaCO_3$)	mg/L	200 (600 max)	98	142	176	Compliant
Chloride	mg/L	250 (1000 max)	24	46	71	Compliant
Nitrate (as NO_3)	mg/L	45	2.1	6.8	12.4	Compliant
Fluoride	mg/L	1.0 (1.5 max)	0.18	0.42	0.71	Compliant
Free Residual Chlorine	mg/L	0.2 - 1.0	0.22	0.46	0.88	Compliant
Arsenic	mg/L	0.01	<0.002	0.003	0.006	Compliant
Lead	mg/L	0.01	<0.001	<0.001	0.003	Compliant
Iron	mg/L	0.3 (1.0 max)	<0.05	0.09	0.21	Compliant

Standards shown are illustrative and based on commonly referenced guidelines (e.g., BIS IS 10500 / WHO). Values are sample data.

3. Microbiological Results

Test	Standard	Samples	Detections	Compliance
E. coli (treated water)	Absent / 100 mL	624	0	100%
Total coliforms (distribution)	Absent / 100 mL	480	0	100%
Heterotrophic plate count	≤ 500 CFU/mL	144	0 above limit	100%

4. Monthly Compliance Trend



The October dip reflects elevated turbidity during heavy rainfall; corrective actions (increased coagulant dosing and filter backwashing) restored normal levels.

5. Sampling Locations Summary

Location Type	Sites	Frequency	Parameters Tested
Source water (reservoir / boreholes)	3	Weekly	Turbidity, pH, TDS, nitrate, arsenic
Treatment plant outlet	1	Daily	Residual chlorine, turbidity, E. coli
Distribution network	18	Weekly	Residual chlorine, coliforms, pH
Consumer taps (random)	40	Monthly	Full panel incl. metals

6. Observations and Recommendations

- **Source protection:** Continue catchment monitoring; nitrate levels are within limits but trending slightly upward (+8% vs. 2024).
- **Monsoon readiness:** Install online turbidity analysers at filter outlets and pre-position additional coagulant stock before the rainy season.
- **Network maintenance:** Prioritise flushing of dead-end mains in Zones 4 and 7 to maintain residual chlorine above 0.2 mg/L.
- **Household guidance:** Residents may consume tap water safely; those with storage tanks should clean them every six months.

7. Contact

Riverbend Water Quality Laboratory | Phone: +00 000 000 0000 | Email: quality@example.org

This is a sample report with fictitious data created for illustration and template purposes only.